

ZENELAIT LMS

COMPREHENSIVE ROLE-BASED LEARNING MANAGEMENT MANUAL

An all-inclusive, multi-role digital learning platform designed for modern educational institutions and training academies. Built by Zenelait InfoTech Private Limited to streamline workflows from classroom attendance to board-level financial and academic audits.

CORPORATE PROFILE

Zenelait InfoTech is a leading software engineering firm dedicated to building next-generation enterprise applications. We specialize in designing customizable SaaS platforms that integrate CRM, ERP, and learning management networks into unified hubs.

ZENELAIT LMS VISION

Zenelait LMS enables educational institutions to achieve 100% operational transparency. Our platform bridges the gap between daily classroom metrics and executive administration auditing, delivering automated compliance checks, integrated payment gateways, and AI-driven student performance insights.

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Security Note:

User role configurations are validated dynamically at database query compile-time. Any route modifications are blocked by client side Auth guards.

1. SYSTEM ARCHITECTURE & TECHNICAL STACK

Zenelait LMS employs a highly responsive, decoupled client-server architecture model. The client-side dashboard framework runs on a React Single Page Application (SPA), connecting to a multi-threaded Spring Boot 3 enterprise application that orchestrates business logic and communicates directly with external cloud-based Generative AI models.

TECHNICAL STACK DETAILS

LAYER	TECHNOLOGY	RESPONSIBILITY
Frontend Client	React 18, Styled-Components, Recharts	Role Dashboards, Real-time Charts, Responsive Layouts, Client-side routing, Export Utilities.
Java Backend	Spring Boot 3.2, Java 17/20, Hibernate JPA, Apache Maven	Business Logic, Multi-role REST Data APIs, Security Filter Chains, and Batch Scheduling.
Generative AI Integration	Google Gemini API (gemini-2.5-flash), Hugging Face Inference API	Intelligent Role-Specific Assistant (student performance, parent audits, attendance, and fee querying).
Database	MySQL 8.x	Relational Schemas, User Ledgers (single-table User model with polymorphic roles), Batches/Courses Catalog, Fee Logs.

2. SECURITY FILTER CHAIN & DECOUPLING MODEL

All network interactions use JSON-encoded requests. Spring Security handles endpoint protection, verifying JWT signature keys encoded in client-header HTTP bearer authentication tokens (`Authorization: Bearer <accessToken>`) on every dispatch through [SecurityConfig](#).

The backend employs database connection pools (HikariCP) to interact with relational schemas, ensuring fast transactional speeds during heavy academic logs updates, attendance submissions, and fee updates.

- Password storage:** Encrypted using BCrypt (strength 10) in the service layer before writing.
- JWT Claims mapping:** Custom filters map roles (`ROLE_ADMIN`, `ROLE_TEACHER`, `ROLE_STUDENT`, `ROLE_PARENT`, `ROLE_ULTRA_SUPER_ADMIN`) mapping controller security mappings.

3. LMS ROLE HIERARCHIES & PERMISSIONS

The platform enforces strict authentication boundaries across 6 distinct user roles:

ROLE	DATABASE FLAG MAPPING	OPERATIONAL CONTEXT PERMISSIONS
ULTRA SUPER ADMIN	ULTRA_SUPER_ADMIN	Full system owner. Manages organizations (tenants), global feature flags, subscription packages, and system-wide revenue forecasts.
SUPER ADMIN	ADMIN (superAdmin = true)	Academy owner. Manages organization-level admins, billing subscriptions, revenue reports, features, courses, and departments.
ADMIN	ADMIN (superAdmin = false)	Operational coordinator. Onboards teachers/students/parents, manages coursework catalogs, publishes announcements, logs fees.
TEACHER	TEACHER	Classroom executor. Publishes assignments, registers daily student attendance logs, evaluates and grades submissions.
STUDENT	STUDENT	Academic workspace. Submits assignments, requests course enrollments, prints verified course completion certificates.
PARENT	PARENT	Academic auditor. Links children using student codes, audits grades/attendance, manages parent wallet for Razorpay payments.

DATABASE POLYMORPHISM LAYOUT

Polymorphic role allocation is mapped using a single-table strategy in Hibernate (users table). User classes are loaded dynamically based on username/email credentials parsed by Spring Security filters.

4. ULTRA SUPER ADMIN DASHBOARD

The Ultra Super Admin Portal (routed at /ultra-super-admin) controls system-wide billing, tenants, features, and platform analytics.

OVERVIEW PAGE (PLATFORM DASHBOARD)

Loads stats via `GET /api/ultra-super-admin/stats` mapping counts:

- **Stats Cards:** Total Organizations, Active Organizations, Total Super Admins, Total Registered Admins, Total Students, Teachers, and Parents system-wide.
- **Quick Action buttons:**
 - *Create Organization:* Launches organization modal.
 - *Create Super Admin:* Launches super admin onboard modal.
 - *View Users by Org:* Switches tab to organization directory.

ORGANIZATIONS MANAGEMENT PAGE

Displays all onboarded academies in a detailed grid via `GET /api/ultra-super-admin/organizations`.

- **Columns:** Academy Name, Email, City, Organization Code, Active Plan, Active Status.
- **Create Button:** Launches creation dialog. Submits form inputs (Name, Email, Phone, Address, City, Country, Description) via `POST /api/ultra-super-admin/organizations`.
- **Status Toggle button (Disable/Enable):** Calls `PATCH /api/ultra-super-admin/organizations/{id}/toggle-active` to instantly disable or enable all login routes and APIs for that academy's users.

USERS BY ORG MANAGEMENT DIRECTORY

Permits the platform owner to inspect the operational directory inside any active tenant organization:

- **Data Collection:** Selecting an active organization triggers calls to:
 - `GET /api/ultra-super-admin/organizations/{id}/students`
 - `GET /api/ultra-super-admin/organizations/{id}/teachers`
 - `GET /api/ultra-super-admin/organizations/{id}/parents`
- **Information Display:** Summarizes counts for Students, Teachers, and Parents. Displays lists containing: User Name, System Code/ID, Email, Phone, and Active Status flag.

GRANULAR FEATURE MANAGEMENT & REVENUE COCKPIT

- **Global System Features:** Create and toggle platform modules globally using `POST /api/ultra-super-admin/features`.
- **Tenant Configuration:** Toggles specific global features for a selected organization via `POST /api/ultra-super-admin/organizations/{orgId}/features/{featureId}/toggle`.
- **Revenue Cockpit:** Fetches metrics via `GET /api/ultra-super-admin/revenue`. Displays MRR, ARR, and charts. Allows MoM target projections simulator slider adjustment.

5. SUPER ADMIN PORTAL (ACADEMY OWNER)

The Super Admin role (authenticated under standard login with `superAdmin = true`) represents the local owner of a tenant academy.

EXCLUSIVE CONTROL VIEWS

- **Admin Management tab:** Accesses local list of administrators. Featuring the "+ Add Admin" button, which registers regular administrators associated with their organization ID via `POST /api/admin/users`.
- **Subscription status tab:** View active package assigned by Ultra Super Admin, showing duration dates and remaining usage days.
- **Local Revenue Report tab:** Generates graphs mapping tuition fee collections across local batches.

6. ADMIN PORTAL (OPERATIONAL COORDINATOR)

Regular administrators (authenticated with `superAdmin = false`) coordinate academic logistics. They lack access to subscription packages, billing, and cannot create other administrator accounts.

OPERATIONAL PAGES & DASHBOARD ACTIONS

- **Users Directories:** Tabbed tables showing Teachers, Students, and Parents.
 - **Actions:** Toggles active status via `PATCH /api/admin/users/{id}/activate` or deletes accounts via `DELETE /api/admin/users/{id}`.
- **Course Catalog:** Adds academic subjects assigning titles, descriptions, and teachers via `POST /api/admin/courses`.
- **Batch Configurator:** Links student registries to academic batches.
- **Fee Logger:** Populates invoice ledgers. Launches fee assignments specifying amount, student ID, and due dates via `POST /api/admin/fees`.
- **Announcements:** Broadcaster tool to send alerts via `POST /api/admin/announcements`.

7. TEACHER PORTAL (ACADEMIC CONTROLS)

The Teacher Portal integrates daily class management tools, attendance registers, and assignments pipelines.

DASHBOARD PAGES & CONTROLS

- **My Courses view:** Fetches courses assigned to the teacher's profile via `GET /api/teacher/courses`.
- **Attendance Log:** Registers student attendance status. Clicking "Submit Attendance" maps student lists and statuses ('PRESENT'/'ABSENT'/'LATE') to the database via `POST /api/teacher/attendance`.
- **Assignments Hub:**
 - *Actions:* Publishes coursework templates using `POST /api/teacher/assignments`.
- **Grading & Feedback:** Teacher reviews student submissions. Submitting marks and feedback notes records the evaluations via `PATCH /api/teacher/submissions/{id}/grade`.

8. STUDENT PORTAL (LEARNING WORKSPACE)

The Student Portal aggregates dashboard elements representing coursework timelines and evaluations.

DASHBOARD PAGES & CONTROLS

- **My Courses view:** Lists enrolled subjects and timetables.
- **Course Enrollment request:** Submits requests to enroll in new batches.
- **Assignments hub:** View assignments lists and grade responses. Clicking "Submit Assignment" uploads files or text answers using `POST /api/student/assignments/{id}/submit`.
- **Completions & Certificates:** Upon course completion, the student selects "Generate Certificate" which compiles student metrics into verified PDF documents using `html2canvas` and `jspdf`.

9. PARENT PORTAL (CHILD AUDITING & WALLET)

The Parent Portal links parents to the learning space to keep track of their children's activities and manage tuition fees.

DASHBOARD PAGES & CONTROLS

- **Children link:** Parents register their children by inputting a student code via `POST /api/parent/children/link`.
- **Audit Dashboard:** Tracks child attendance data via `GET /api/parent/children/{childId}/attendance` and tuition logs via `GET /api/parent/children/{childId}/fees`.
- **Integrated Wallet:** Links tuition records directly to a wallet ledger.
 - *Wallet Top-up:* Reloads parent account funds using the integrated Razorpay Gateway API.
 - *Invoice Settlement:* Parents pay tuition fees by clicking "Settle Bill". This subtracts funds from the wallet and marks the invoice as PAID in the database.

10. AI ASSISTANT INTEGRATION (GEMINI PIPELINE)

Zenelait LMS features direct API integration with Google Gemini 2.5 Flash and Hugging Face Inference servers to provide contextual assistance based on active database records via `POST /api/ai/chat`.

PROMPT CONTEXT COLLECTION PIPELINE

When a user initiates a query, the backend executes context collection algorithms based on their role:

- **Student Context:** Gathers class grades, attendance sheets, and enrolled courses before calling the LLM.
- **Parent Context:** Pulls linked children's details, attendance ratios, and outstanding tuition balances.
- **Teacher Context:** Compiles curriculum plans and student feedback logs.
- **LLM Forwarding:** Formulates a structured system prompt, appends user context, and posts data directly to the Gemini API endpoint.